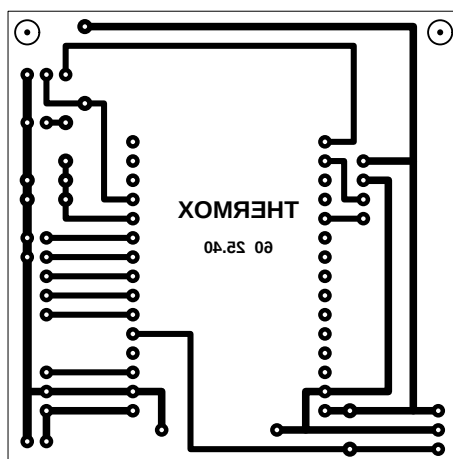
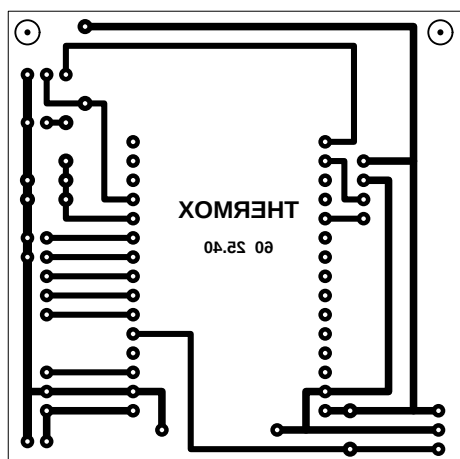
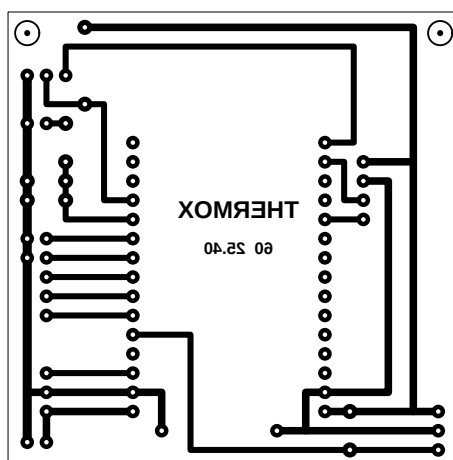
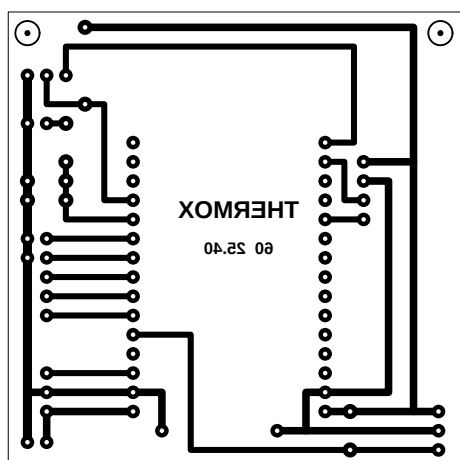


ThermoX board 60x60 mm - COPPER for toner transfer - ESP32 rows 25.40 mm

Use **THIS** page if the two pin rows of your ESP32 are 25.40 mm (1.0 in) apart, centre to centre.

If yours are 22.86 mm apart, use the other copper page. Easiest check: lay the ESP32 on the board drawing below.

LASER printer, 100% / 'Actual size' (turn OFF 'Fit to page'), glossy or photo paper. Do NOT mirror: the small text reads BACKWARDS here and will read normally on the copper. 4 copies, pick the cleanest one.



SCALE CHECK: this bar must measure exactly 50 mm.

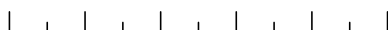
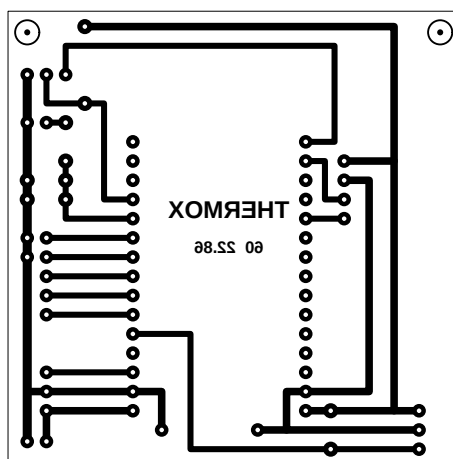
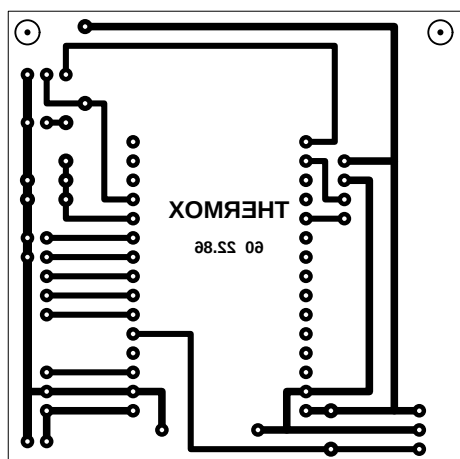
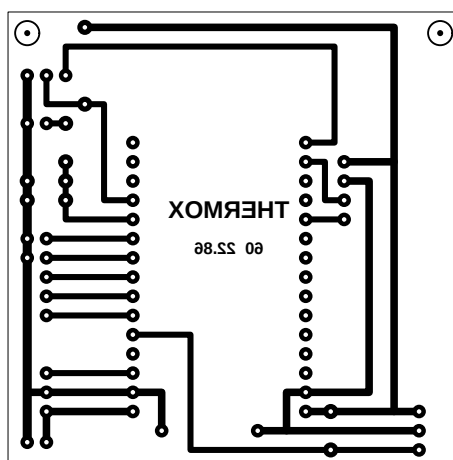
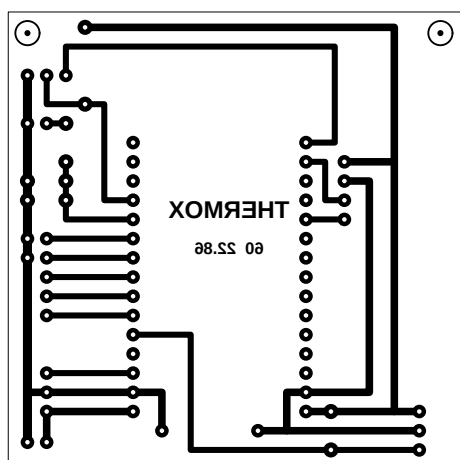
Board 60.0 x 60.0 mm, cut on the outer line. White dots = drill centres (pins 0.8-1.0 mm, terminal 1.2 mm, holes 3.2 mm).

ThermoX board 60x60 mm - COPPER for toner transfer - ESP32 rows 22.86 mm

Use **THIS** page if the two pin rows of your ESP32 are 22.86 mm (0.9 in) apart, centre to centre.

If yours are 25.40 mm apart, use the other copper page. Easiest check: lay the ESP32 on the board drawing below.

LASER printer, 100% / 'Actual size' (turn OFF 'Fit to page'), glossy or photo paper. Do NOT mirror: the small text reads BACKWARDS here and will read normally on the copper. 4 copies, pick the cleanest one.



10 x 10 mm square

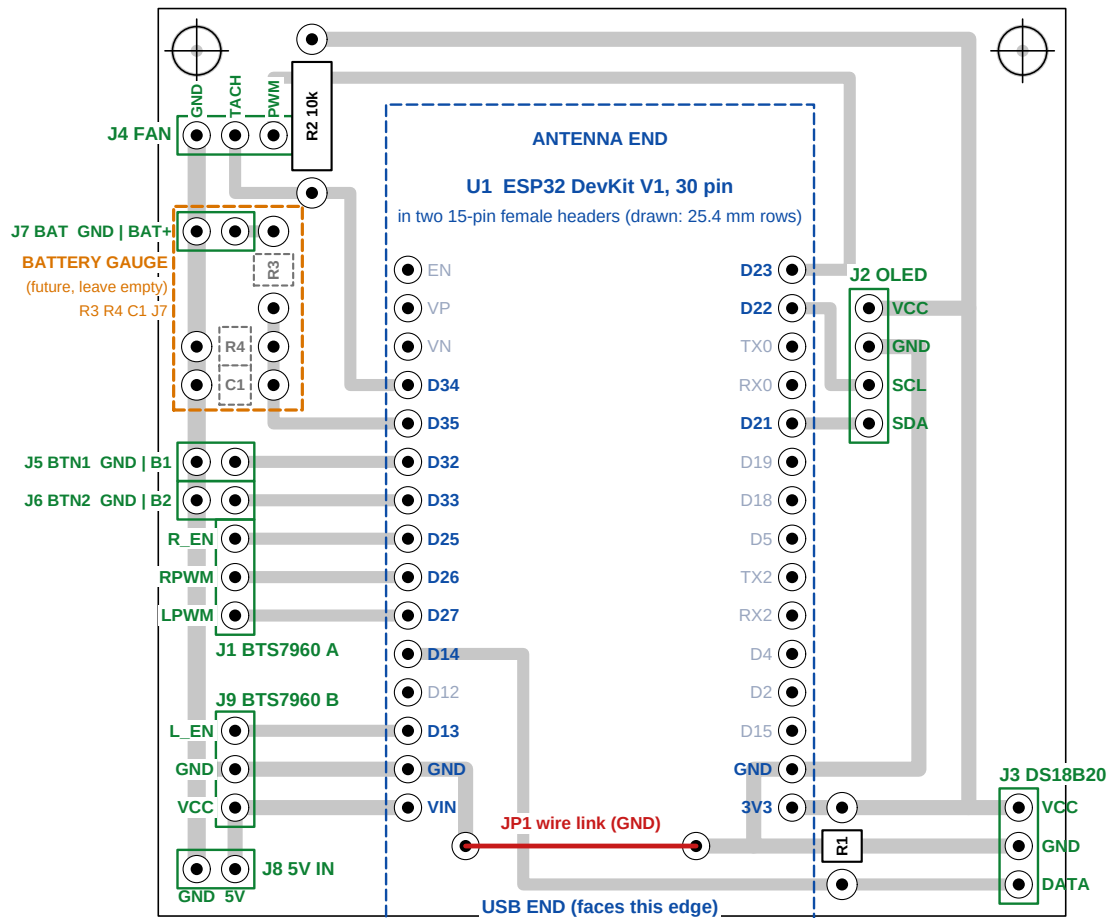
SCALE CHECK: this bar must measure exactly 50 mm.

Board 60.0 x 60.0 mm, cut on the outer line. White dots = drill centres (pins 0.8-1.0 mm, terminal 1.2 mm, holes 3.2 mm).

Parts placement and drilling guide (TOP view, 2x size, not for transfer)

Parts go on the plain (non-copper) side. Grey = copper on the bottom, seen through the board.

Drawn for 25.4 mm ESP32 rows. On the 22.86 mm board the right socket row is one pin space (2.54 mm) closer; all other parts are the same.



J4 fan: black GND, yellow FG, blue PWM (red +12V goes to the 12V supply). J3 DS18B20: red VCC, black GND, yellow DATA.
J7 (battery gauge) is for later: leave it open. Power wires enter J8 from the bottom edge.

Order of assembly

1. JP1 wire link first (a cut resistor leg). It sits under the ESP32.
2. R1 4.7k and R2 10k. Leave R3, R4 and C1 EMPTY (future battery gauge).
3. Male pin headers J1-J6, J7 and J9 (2.54 mm), then the J8 screw terminal.
4. The two 15-pin FEMALE headers last. Plug the ESP32 in with VIN/GND at the bottom-left, 3V3/GND at the bottom-right and the USB port toward the bottom edge.

Drill sizes

Header and resistor pins 0.8-1.0 mm | screw terminal 1.2-1.3 mm | mounting holes 3.2 mm (M3)

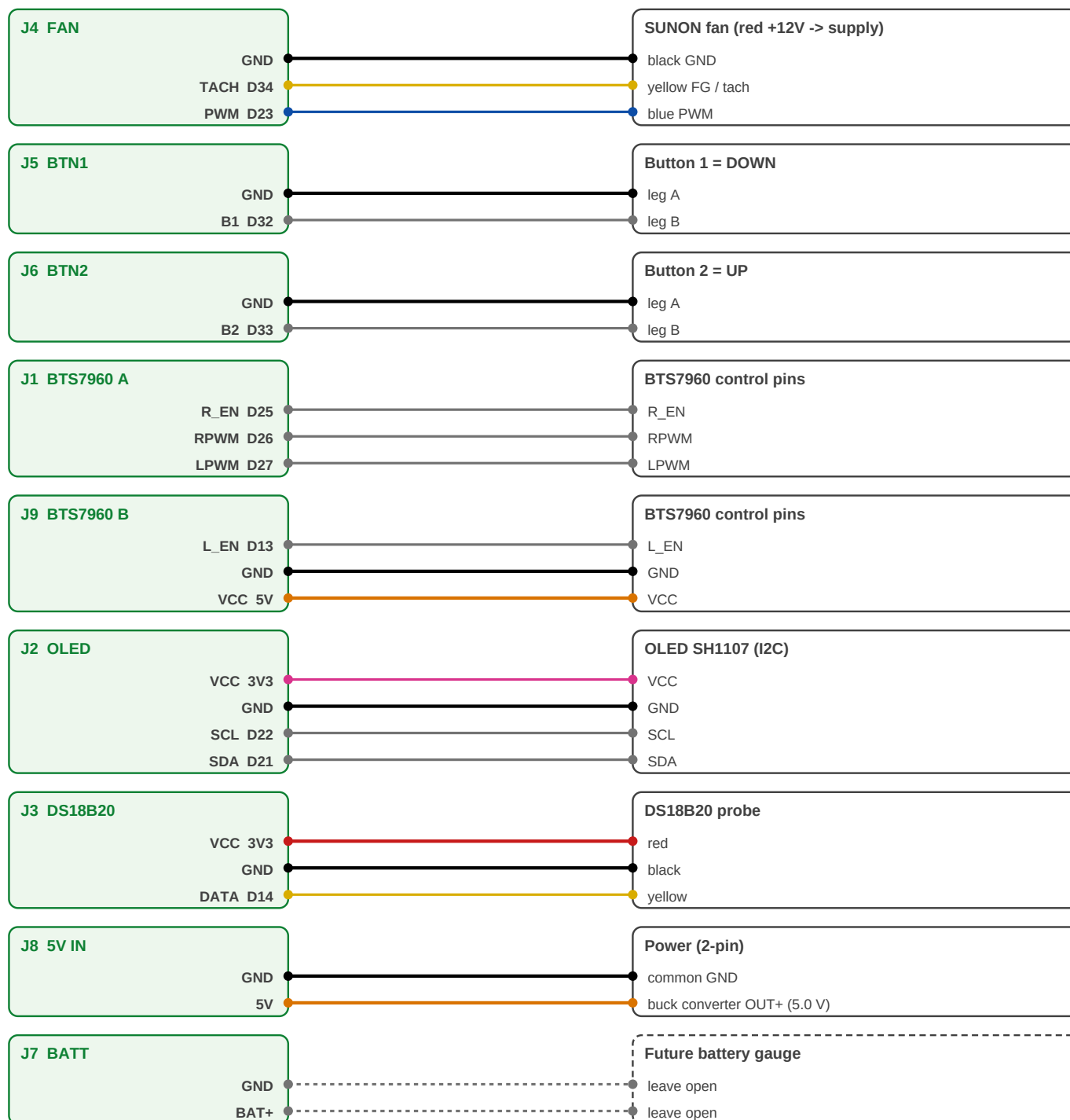
Check before you solder the ESP32 headers

No copper bridges between neighbouring pads (scrape with a blade, test with a multimeter).

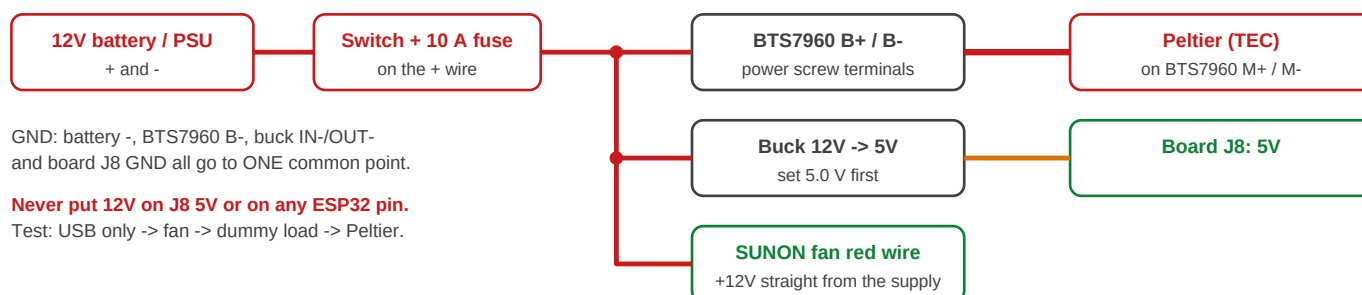
5V, 12V and 3V3 must NOT beep to GND on the continuity test.

Wiring (line route): board connectors to ThermoX parts

Female-female dupont wires on the pin headers. Match the pin NAMES printed on each module.



Power path (Peltier current never goes through the board)



Parts list, GPIO check and how to make the board

Parts

Ref	Part	Use
U1	ESP32 DevKit V1, 30 pin	+ two 15-pin female headers, 2.54 mm
J1, J9	3-pin male header (x2)	BTS7960: R_EN RPWM LPWM / L_EN GND VCC
J2	4-pin male header	OLED: VCC(3V3) GND SCL SDA
J3	3-pin male header	DS18B20: VCC(3V3) GND DATA
J4	3-pin male header	SUNON fan: GND TACH PWM (red wire to 12V)
J5, J6	2-pin male header (x2)	buttons: GND + signal
J8	2-pin male header (or solder wires)	power in: GND 5V from the buck
R1	4.7 kohm, 1/4 W	DS18B20 DATA pull-up to 3V3
R2	10 kohm, 1/4 W	fan tach pull-up to 3V3 (GPIO34 has none)
JP1	wire link (cut resistor leg)	joins left and right GND
PCB	single-sided copper clad	at least 64 x 64 mm
	BATTERY GAUGE (future) - leave these EMPTY now:	
R3	100 kohm, 1/4 W	divider top: BAT+ to GPIO35
R4	100k (1S Li-ion) or 27k (3S 12V)	divider bottom: GPIO35 to GND
C1	100 nF ceramic	smooths the GPIO35 reading
J7	2-pin male header	BAT+ wire from the battery, and GND

GPIO check (current ThermoX pin map, nothing changed)

Function	GPIO	Board pin
Button 1	GPIO32	J5 B1
Button 2	GPIO33	J6 B2
OLED SDA	GPIO21	J2 SDA
OLED SCL	GPIO22	J2 SCL
BTS7960 LPWM	GPIO27	J1 LPWM
BTS7960 RPWM	GPIO26	J1 RPWM
BTS7960 LEN	GPIO13	J9 L_EN
BTS7960 REN	GPIO25	J1 R_EN
Temperature sensor (DS18B20)	GPIO14	J3 DATA (+ R1 4.7k)
SUNON fan PWM	GPIO23	J4 PWM
SUNON fan tach / FG	GPIO34	J4 TACH (+ R2 10k)
Battery sense (future, off)	GPIO35	R3/R4/C1 divider -> J7

Toner transfer, step by step

1. Print page 1 OR page 2 (the one that matches your ESP32 rows) on glossy/photo paper, laser, 100%.
2. Measure the 50 mm scale bar. Put your ESP32 on the print and check that its pins land on the pads.
3. Sand the copper lightly (fine sandpaper or steel wool), clean with alcohol/acetone, don't touch it.
4. Toner side DOWN on the copper. Iron on the hottest (cotton) setting, no steam, 3-5 min, firm pressure, go slowly over every edge and corner.
5. Let it cool, soak in warm water 10-15 min, rub the paper off gently with your thumb.
6. Touch up broken traces with a permanent marker. The text must read NORMALLY on the copper.
7. Etch in ferric chloride (warm, keep it moving). Remove the toner with acetone.
8. Drill at the dots, cut along the outline, solder in the order on page 3.

Before first power-up

- * Continuity: 5V-GND, 12V-GND and 3V3-GND must be OPEN (no beep). No bridges between ESP32 pads.
- * First power: USB only (no 12V, no BTS7960). Check the OLED, both buttons and the temperature.
- * Then 12V + buck 5V, the fan, the BTS7960 with a dummy load, and only then the Peltier.

Notes

- * For the 30-pin DevKit V1 only. Page 1 = rows 25.4 mm apart, page 2 = rows 22.86 mm. 38-pin will NOT fit.
- * Battery gauge (later): fit R3, R4, C1, J7; GPIO35 must stay under 3.1 V. 1S Li-ion: 100k/100k.
3S 12 V pack: 100k/27k and set BATT_DIVIDER = 4.7 in the firmware. Then ENABLE_BATTERY_SENSE.
- * Fan PWM is driven straight from GPIO23, as in the ThermoX firmware README.